PPC-checkpoint-2

Curvature Optimisation

Once the spline is built: replace the boundary conditions with $f^{n+1}(p) = 0$

our goal is to optimize the spline such that the total curvature is minimized for smoother path

There are 3 parts to Optimisation problem:

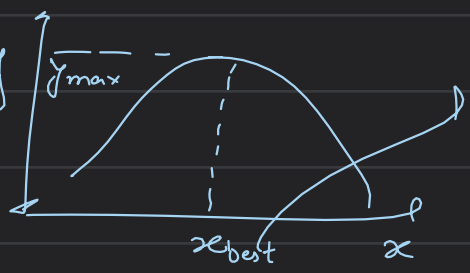
① objective (or cost) func. ↳ our case: minimizing curve

② Parameters (or Decision Variables) ↳ the var. to be changed to get the best value of objective

③ Constraints: rules or limits which the parameters must satisfy

what is optimisation? choosing inputs that will result in best possible output ↳ consist of best soln to maximise or minimise a problem

eg:



this x_{best} is the optimised soln for finding the max y value

⊗ optimisation of input such that you get the best way to get the maximum or minimum output

Objective function: ↳ the value that is to be optimised (in our case curvature of the path)

eg: minimize cost, maximize speed, minimize weight etc

Area = a×b , f(x) = Area

Parameters (Decision Variables): values the optimizer can change to best way optimize f(x), eg: Parameters: a, b

Gradients: slope : Df ↳ by differentiation

Constraints: eg: $a \times b \leq 5$ ↳ a constraint

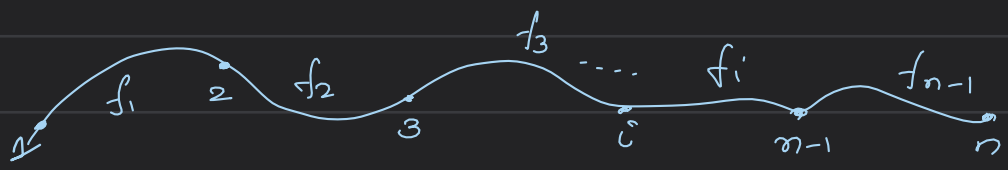
or $a + b = 10$

Method of gradient Descent: we optimize the waypoints to get a new set of waypoints which are then undergone cubic spline interpolation to get a new curve

Step-1: to cubic spline interpolation to get all the polynomials

$f_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$

$i = 1$ to $n-1$ $n-1$: no. of polynomials, n: total no. of waypoints



Step-2: put $x(t) = t$, $y(t) = f(t)$

& compute $|K(t)| = \dfrac{|x''(t)y'(t) - y''(t)x'(t)|}{((x'(t))^2 + (y'(t))^2)^{3/2}}$

$|K(t)|$ ↓ curvature

Step-3: compute total curvature: $K_i(t)$ ↳ curvature of i^{th} polynomial $f_i(x)$

$J = \sum_i (K_i(t))^2$

↳ total curvature

Step-4: now we have to optimise the waypoints to get the minimum total curvature

⇒ slightly move the waypoints to get the minimum curvature

↳ How much?

$y_i = y_i - eta * gradient$

↳ learning rate ⇒ const

$\downarrow$

y_{opt}

$\dfrac{\partial J}{\partial y_i} = \dfrac{J(y_i + \varepsilon) - J(y_i)}{\varepsilon}$

↳ movement

↳ how sensitive the curvature is to waypoint

gradient points towards inc curvature

logic: waypoint with sharp bend → large $\frac{\partial J}{\partial y_i}$

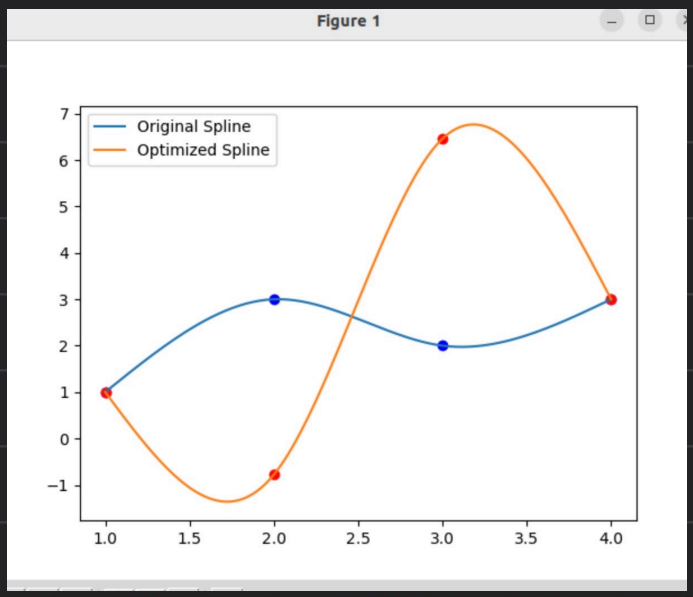
so $y_{new} << y_{old}$ peak flatens

Step-3: we will get a new set of way points $(x, y\text{-opt})$

↓

recompute the spline

Task-1: curvature optimization for 4 points



curvature_optimization > curvature_optimization > gradient_descent.py > main > computeCurvature

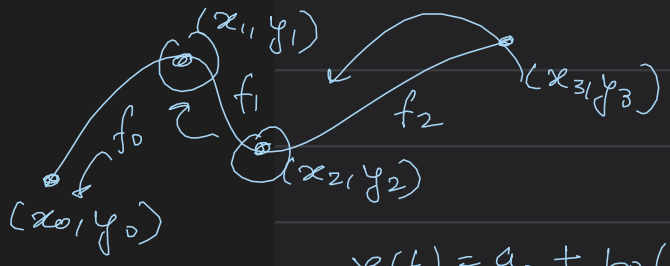
```python
import matplotlib.pyplot as plt
import numpy as np

def main(args=None):
    eta = 0.01
    x = np.array([1, 2, 3, 4])
    y = np.array([1, 3, 2, 3], dtype = float)
    A = np.zeros((12, 12))
    B = np.zeros(12)
    eqn = 0
    for i in range(3):
        A[eqn, 4*i] = 1
        B[eqn] = y[i]
        eqn += 1
        dx = x[i+1] - x[i]
        A[eqn,4*i] = 1
        A[eqn, 4*i+1] = dx
        A[eqn, 4*i+2] = dx**2
        A[eqn, 4*i+3] = dx**3
        B[eqn] = y[i+1]
        eqn += 1
    for i in range(1,3):
        dx = x[i] - x[i-1]

        A[eqn, 4*(i-1)+1] = 1
        A[eqn, 4*(i-1)+2] = 2*dx
        A[eqn, 4*(i-1)+3] = 3*dx**2
        A[eqn,4*i+1] = -1
        eqn += 1
    for i in range(1,3):
        dx = x[i] - x[i-1]
        A[eqn, 4*(i-1)+2] = 2
        A[eqn, 4*(i-1)+3] = 6*dx
        A[eqn, 4*i+2] = -2
        eqn += 1
```

```
    A[eqn, 2] = 2
    eqn += 1
    dx = x[3] - x[2]
    A[eqn, 10] = 2
    A[eqn, 11] = dx*6
    eqn += 1
    coeffs = np.linalg.solve(A, B)
```

```
def computePoly(i, xi):
    a = coeffs[4*i]
    b = coeffs[4*i+1]
    c = coeffs[4*i+2]
    d = coeffs[4*i+3]
    return a + b*(xi-x[i]) + c*(xi-x[i])**2 + d*(xi-x[i])**3

X, Y = [], []
for i in range(3):
    xi_vals = np.linspace(x[i], x[i+1], 50)
    yi_vals = computePoly(i, xi_vals)
    X.extend(xi_vals)
    Y.extend(yi_vals)
```

→ got the cubic spline

```
def computeCurvature(y_input):

  B_temp = np.zeros(12)

  eqn = 0

  for i in range(3):

    B_temp[eqn] = y_input[i]
    eqn += 1

    B_temp[eqn] = y_input[i+1]
    eqn += 1

  B_temp[eqn] = 0
  eqn += 1
  B_temp[eqn] = 0

  coeffs_temp = np.linalg.solve(A, B_temp)

  total_curvature = 0
```



$$y_0(t) = a_0 + b_0(t-x_0) + c_0(t-x_0)^2 + d_0(t-x_0)^3$$

$$y_{new}\big|_0 = y\big|_0 - \eta \frac{dJ}{dy}$$

↓

got new points

↓

new polynomial

```
  for i in range(3):

    xi_vals = np.linspace(x[i], x[i+1], 50)

    for xi in xi_vals:

      b = coeffs_temp[4*i+1]
      c = coeffs_temp[4*i+2]
      d = coeffs_temp[4*i+3]

      yd = b + 2*c*(xi - x[i]) + 3*d*(xi - x[i])**2
      y2d = 2*c + 6*d*(xi - x[i])

      kappa = abs(y2d)/(1 + yd**2)**1.5

      total_curvature += kappa**2

  return total_curvature

J = computeCurvature(y)
```

$$\frac{dJ}{dy} : \frac{J(y+\varepsilon) - J(y)}{\varepsilon}$$

① compute J for y

② compute J for $(y+\varepsilon)$

↓

$(x_i, y_i + \varepsilon)$

↓

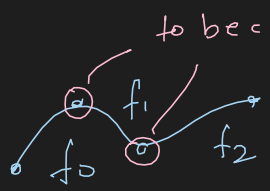
new set of points

↓

the polynomial changes → new a, b, c, d

→ for a particular y

```
def computeGradient(index):
  eps = 0.000001
  J1 = computeCurvature(y)
  y_temp = y.copy()
  y_temp[index] += eps
  J2 = computeCurvature(y_temp)

  print(J1)
  print(J2)
  print(J2 - J1)

  gradient = (J2 - J1)/eps
  return gradient

grad1 = computeGradient(1)
grad2 = computeGradient(2)

print(grad1)
print(" ")
print(grad2)
print("\n")
```



for $\int$ of $y+\varepsilon$

the total curvature will be calculated using the new polynomial of $(x_i, y_i+\varepsilon)$

index → for f_1 / for f_2

```
y_opt = y.copy()

y_opt[1] -= eta*grad1
y_opt[2] -= eta*grad2

B_opt = np.zeros(12)
```

all points

} we are moving non-end points

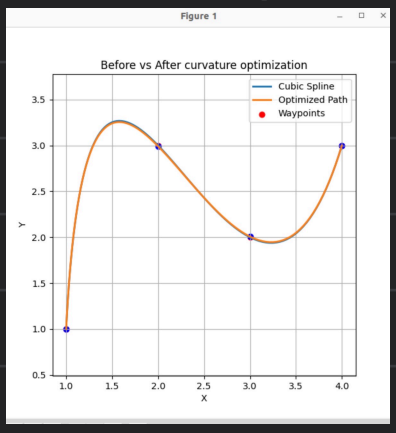
```
eqn = 0
for i in range(3):

  B_opt[eqn] = y_opt[i]
  eqn += 1

  B_opt[eqn] = y_opt[i+1]
  eqn += 1

coeffs_opt = np.linalg.solve(A, B_opt)

X_opt = []
Y_opt = []

for i in range(3):

  xi_vals = np.linspace(x[i], x[i+1], 50)

  a = coeffs_opt[4*i]
  b = coeffs_opt[4*i+1]
  c = coeffs_opt[4*i+2]
  d = coeffs_opt[4*i+3]

  yi_vals = (
   a
   + b*(xi_vals-x[i])
   + c*(xi_vals-x[i])**2
   + d*(xi_vals-x[i])**3
)
  X_opt.extend(xi_vals)
  Y_opt.extend(yi_vals)
```

→ new coeff of the poly from the optimised y-coord

} new set of values for plotting of the new cubic spline after optimization

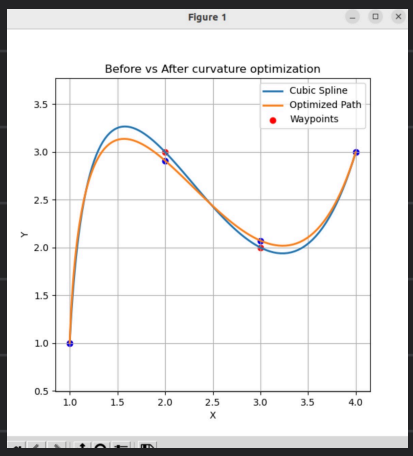
```
plt.plot(X, Y, label="Original Spline")
plt.plot(X_opt, Y_opt, label="Optimized Spline")
plt.scatter(x, y, color='blue')
plt.scatter(x, y_opt, color='red')
plt.legend()
plt.show()
```

Task-2: Curvature Optimization using libraries



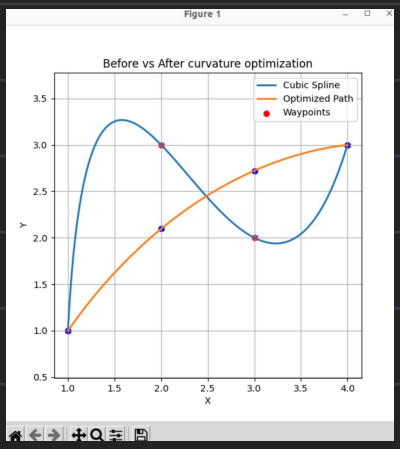
eta = 0.001

eps = 0.00001



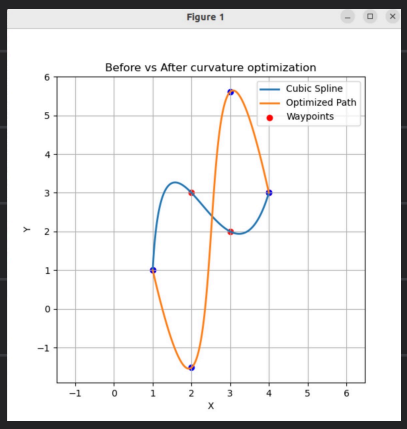
eta = 0.01

eps = 0.0001



eta = 0.1

eps = 0.001



eta → 1

eps = 0.01

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.interpolate import splprep, splev
from scipy.optimize import approx_fprime

def main(args=None):
    eta = 0.001
    x = np.array([1, 2, 3, 4])
    y = np.array([1, 3, 2, 3], dtype=float)



    tck, u = splprep([x, y], s=0, per=False)
    u_new = np.linspace(0, 1, 100)
    x_smooth, y_smooth = splev(u_new, tck)

    def computeCurvature(y_opt):
        y_full = np.concatenate([[y[0]], y_opt, [y[-1]]])
        tck_, _ = splprep([x, y_full], s=0, per=False)
        dx, dy   = splev(u_new, tck_, der=1)
        d2x, d2y = splev(u_new, tck_, der=2)
        kappa_sq = (dx * d2y - dy * d2x)**2 / (dx**2 + dy**2)**3
        ds = np.sqrt(dx**2 + dy**2)
        return np.trapz(kappa_sq * ds, u_new)

    y_opt = y[1:-1].copy()

    eps = 0.00001
    gradient = approx_fprime(y_opt, computeCurvature, eps)

    y_opt = y_opt - eta * gradient
    y_opt = np.concatenate([[y[0]], y_opt, [y[-1]]])

    tck_opt, _ = splprep([x, y_opt], s=0, per=False)
    x_opt_smooth, y_opt_smooth = splev(u_new, tck_opt)

    plt.figure(figsize=(6, 6))
    plt.plot(x_smooth, y_smooth, label='Cubic Spline', linewidth=2)
    plt.plot(x_opt_smooth, y_opt_smooth, label="Optimized Path", linewidth = 2)
    plt.scatter(x, y, color='red', label='Waypoints')
    plt.scatter(x, y_opt, color='blue')
    plt.xlabel("X")
    plt.ylabel("Y")
    plt.title("Before vs After curvature optimization")
    plt.legend()
    plt.grid()
    plt.axis('equal')
    plt.show()
```

(*) these four graphs shows that the value of eta should neither very large nor to small

compute t,c,k

input t,c,k & get the cubic spline

total curvature

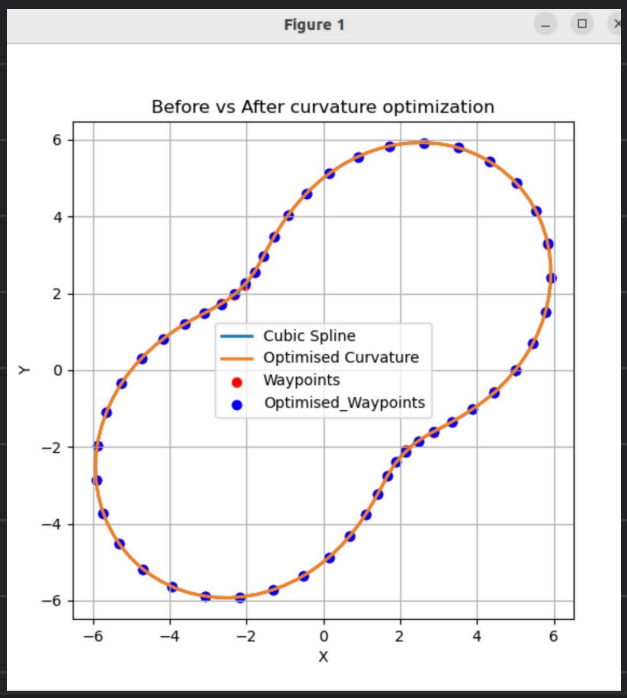
y-opt is only the non-end points which will be changed

fxn to compute gradient

initially y-opt was only non-end points but after curvature optimization, y-opt is changed to the entire set of y-coord by concatenation with y[0] & y[-1]

computing the new cubic spline from the final y-opt

Task. 3: Curvature optimisation of csv using libraries




```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.interpolate import splprep, splev
from scipy.optimize import approx_fprime
import os

def main(args=None):
    eta = 0.01
    csv_path = os.path.expanduser('/home/harshita/Downloads/loop_track_waypoints.csv')

    dataPoints = pd.read_csv(csv_path)
    x = dataPoints['X'].values
    y = dataPoints['Y'].values

    if (x[0] != x[-1]) or (y[0] != y[-1]):
        x = np.append(x, x[0])
        y = np.append(y, y[0])

    tck, u = splprep([x, y], s=0, per=True)
    u_new = np.linspace(0, 1, 100)
    x_smooth, y_smooth = splev(u_new, tck)

    def computeCurvature(y_opt):
      tck_, _ = splprep([x, y_opt], s=0, per=True)
      dx, dy = splev(u_new, tck_, der=1)
      d2x, d2y = splev(u_new, tck_, der=2)

      kappa = np.abs(dx*d2y - dy*d2x) / (dx**2 + dy**2)**1.5
      J = np.sum(kappa**2)
      return J

    y_opt = y.copy()#considerng non-endpoints
    eps = 0.00001

    gradient = approx_fprime(y_opt, computeCurvature, eps)
y_opt -= eta*gradient
y_opt[0] = y[0]
y_opt[-1] = y[-1]

tck_opt, u_opt = splprep([x, y_opt], s=0, per = True)
x_opt_smooth, y_opt_smooth = splev(u_new, tck_opt)

plt.figure(figsize=(6, 6))
plt.plot(x_smooth, y_smooth, label='Cubic Spline', linewidth=2)
plt.plot(x_opt_smooth, y_opt_smooth, label = "Optimised Curvature", linewidth = 2)
plt.scatter(x, y, color='red', label='Waypoints')
plt.scatter(x, y_opt, color = 'blue', label = 'Optimised_Waypoints')
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Before vs After curvature optimization")
plt.legend()
plt.grid()
plt.axis('equal')
plt.show()
```

read x from X column & y from Y column

compute the non-optimized polynomial

returns the total curvature

Now here y_opt is firstly all the points including the end points

firstly all the points are changed

then the endpoints are made same as before

new cubic spline after optimization

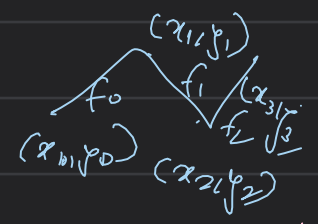
same scale of x & y axes

$$f_i(x) = a + b(x-x_i) + c(x-x_i)^2 + d(x-x_i)^3$$

$$f_0(x) = a + b(x-x_0) + c(x-x_0)^2 + d(x-x_0)^3 \qquad i = 0,1,2$$

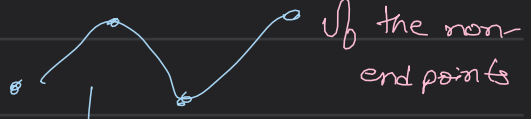
$$f_1(x) = a + b(x-x_1) + c(x-x_1)^2 + d(x-x_1)^3$$

$$f_2(x) = a + b(x-x_2) + c(x-x_2)^2 + d(x-x_2)^3$$



$y \rightarrow y\text{-opt}$ ① Compute curvature by gradient descent

(*) we only change the y coord of the non-end points



we have to change

$$f = [y_0, y_1, y_2, y_3]$$

$y_i = f_i(x)$ $\quad J = \left(y - \eta \frac{\partial J}{\partial y}\right)(\eta)$ $\quad y\text{-opt}$

$$J = \sum |K|^2 \qquad K_i = \frac{|y_i''(x)|}{(1+(y_i'(x))^2)^{3/2}}$$

$y\text{-opt}$ such that the curvature is optimized

$$J = \left(\frac{y_0''(x)}{(1+y_0(x)^2)^{3/2}}\right)^2 + \left(\frac{y_1''(x)}{(1+(y_1(x))^2)^{3/2}}\right)^2 + \left(\frac{y_2''(x)}{(1+(y_2(x))^2)^{3/2}}\right)^2$$

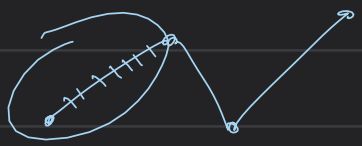
① graph-1 : $(x, y) \rightarrow (x, y)$

② graph-2 : $(x, y\text{-opt}) \rightarrow (x, y\text{-opt})$

$$J = K_0^2 + K_1^2 + K_2^2$$

$$(y\text{-opt})_0 = y_0 - \frac{(J(y_0+\eta) - J(y_0))}{\eta}\eta$$

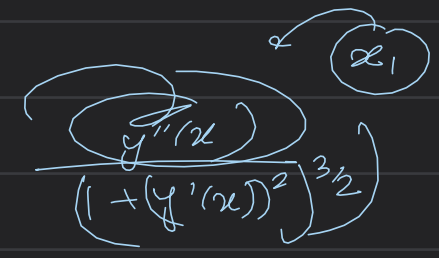
$$(y\text{-opt})_1 = y_1 - \frac{(J(y_1+\eta) - J(y_1))}{\eta}\eta$$



$$(y\text{-opt})_2 = y_2 - \frac{(J(y_2+\eta) - J(y_2))}{\eta}\eta$$

K_i curvature at any point (x_i, y_i)

$$Y\text{-opt} = [(y\text{-opt})_0, (y\text{-opt})_1, (y\text{-opt})_2]$$



$$K_1(x) = \left(\frac{y''(x)}{(1+(y'(x))^2)^{3/2}}\right)$$

$$Y = [y_0, y_1, y_2]$$

↓ from cubic spline interpolation

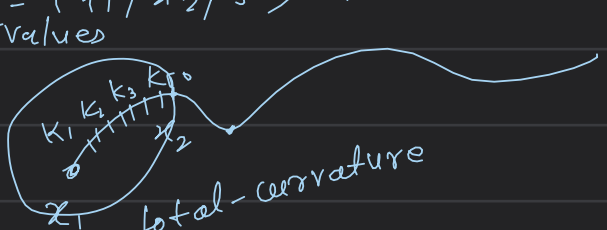
$$y'(x) = b + 2c(x - x_i) + 3d(x - x_i)^2$$

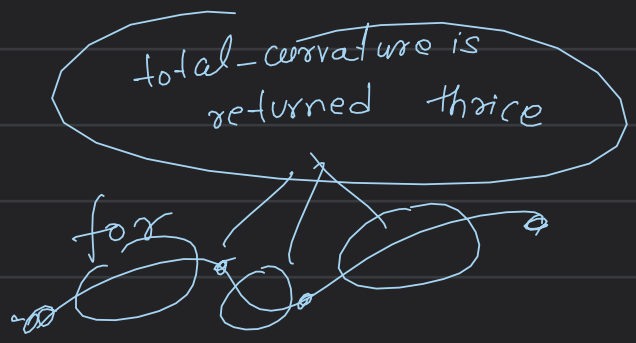
$$y''(x) = 2c + 6d(x - x_i)$$

for waypoint (x_1)

$$k_i(x) = \frac{2c + 6d(x - x_i)}{\left(1 + \left(b + 2c(x - x_i) + 3d(x - x_i)^2\right)^2\right)^{3/2}}$$

$x_i = (x_1, x_2, \ldots)$ values





J = compute curvature (y)

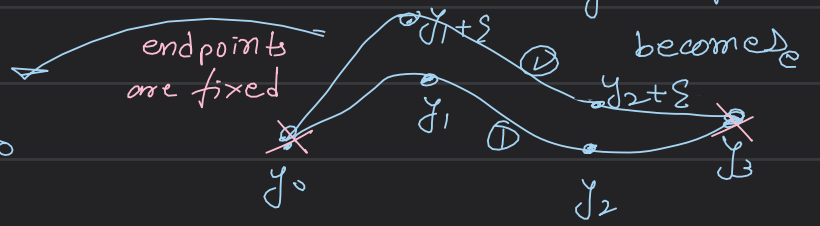
↓

compute total curvature of y_0

$$y_{-opt} \rightarrow y_0 - \left(\eta \frac{\partial J}{\partial y}\right) \rightarrow \frac{J \text{ at } y_0 + \varepsilon \; - \; J \text{ at } y_0}{\varepsilon}$$

y_{-opt} ↓ new choosen waypoint

So on changing y y_0 to $(y_0 + \varepsilon)$ our original splin becomes



So for computation of $\frac{\Delta J}{\Delta y}$ we are suppose to again find new coeff

so that we can find new $y'(x)$ & $y''(x)$ thus new k_i

thus new $J \rightarrow J'$

$$\frac{\partial J}{\partial y} = \frac{J' - J}{\varepsilon}$$

① is the original spline,

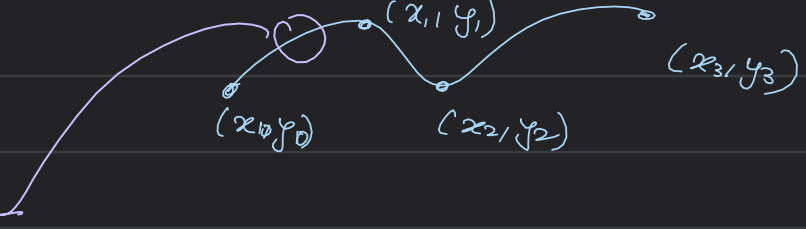
② is just a slightly diff spline created just to compute $\frac{\partial J}{\partial y}$

now we will find $y_{-opt} = y - \eta \frac{\partial J}{\partial y}$

for a particular waypoint

↓

y_{-opt} → new y-coor of all the waypoints ST the curvature is minimized



② compute gradient:

for say way point ① we have to consider this part of the curve or f_0 & to find total curvature of all the 50 points for this part (J)

Note: curvature is a property of a point $(k_1 \dots k_{50})$

$\propto (x - x_i)$ ↓ x_i : that choosen way point

all 50 points w.r.t

```
def computeGradient(index)
    : is basically
    J1 = computeCurvature(y0)        (1) → 0,1,2
    J'1 = computeCurrature(y0+ε)      ε : no. of gradients to be computed
    Gradient = (J'1 - J1) / ε
    return Gradient
```

this fxn returns 3 Gradients

⊛ Gradient are of total curvature ↓ 3 total curvature ↓ 3 Gradients

so computeGradient(0) → Gradient of

computeGradient(1)

computeGradient(2)

Now, compute y-opt ↓

we will compute y-opt for non-end points ⟹ y[1] & y[2] & not of y[0] & y[3]

```
y-opt[0] = y[0]
y-opt[1] = y[1] - eta * computeGradient(1)
y-opt[2] = y[2] - eta * computeGradient[2]
y-opt[3] = y[3]
```

↳ our set of optimized points ← now we will calculate the spline & thus get the optimized curve